

CCF Database: A Machine-Learning-Annotated Corpus of 266,271 Canadian Climate Articles (1978–2024)

Supplementary Information

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This document collects the Supplementary Information accompanying the CCF Database manuscript. It contains: prior-literature and corpus composition tables (Supplementary Tables S1 and S2), the complete annotation framework with operational definitions (Supplementary Table S3), the complete per-category training metrics and dataset distribution (Supplementary Tables S4–S5), the Named Entity Recognition performance summary (Supplementary Table S6), the per-category validation metrics on the 1,000-sentence gold standard (Supplementary Table S7), the per-category distribution of annotations across the database (Supplementary Table S8), and four additional tables added during revision: per-category inter-coder reliability (Supplementary Table S9), reliability tier assignment (Supplementary Table S10), per-model training provenance (Supplementary Table S11), and the complete data dictionary of the enriched database (Supplementary Table S12).

Reproducibility note. Tables S1, S2, S3, and S6 are static prose / lookup tables inlined directly in this LaTeX source (their content does not depend on the live data). Tables S4 to S12 are reproducibly generated by `Scripts/Annotation/17_generate_tables.py` from the canonical CSVs in the `Database/Training_data/` directory and the live CCF Database (Table S8 only); the generator reuses the category-normalisation API exposed by `Scripts/Annotation/15_normalize_categories.py`. Re-running scripts 15, 16, and 17 followed by a fresh `pdflatex` pass regenerates every numerical value in this document from the underlying data.

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Supplementary Table S1. Previous studies on climate change coverage in Canadian newspapers.

Study	Newspaper(s)	Scope	Language	Time Period
Good [1]	The Globe and Mail	National	English	2007
	The National Post	National	English	
	The Toronto Star	National	English	
	The Calgary Herald	Regional	English	
	The Edmonton Journal	Regional	English	
	The Chronicle Herald	Regional	English	
	The Montreal Gazette	Regional	English	
	The Charlottetown Guardian	Regional	English	
	Halifax Daily News	Regional	English	
	The Ottawa Citizen	Regional	English	
	The Star Phoenix	Regional	English	
	The Telegram	Regional	English	
	The Times Colonist	Regional	English	
	The Vancouver Sun	Regional	English	
	The Winnipeg Sun	Regional	English	
	Yukon News	Regional	English	
Rowe [2]	The Globe and Mail	National	English	2007–2008
	The Toronto Star	National	English	
	The Calgary Herald	Regional	English	
	The Toronto Sun	Regional	English	
	The Montreal Gazette	Regional	English	
	The Ottawa Citizen	Regional	English	
Young and Dugas [3]	The Globe and Mail	National	English	1988–2008
	The National Post	National	English	
Anne DiFrancesco and Young [4]	The Globe and Mail	National	English	2008
	The National Post	National	English	
Ahchong and Dodds [5]	The Globe and Mail	National	English	1988–2007
	The Toronto Star	National	English	
Stoddart and Tindall [6]	The Globe and Mail	National	English	1999–2010
	The National Post	National	English	

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Study	Newspaper(s)	Scope	Language	Time Period
Ford and King [7]	The Globe and Mail The Toronto Star	National National	English English	1993–2013
Stoddart and Smith [8]	The Globe and Mail The National Post	National National	English English	2006–2010
Stoddart et al. [9]	The Globe and Mail The National Post	National National	English English	1997–2010
Barkemeyer et al. [10]	The Globe and Mail The National Post The Toronto Star The Vancouver Sun The Toronto Sun	National National National Regional Regional	English English English English English	2008
King et al. [11]	The Globe and Mail The National Post The Toronto Star The Calgary Herald The Chronicle Herald The Vancouver Sun The Whitehorse Daily Star Journal de Montréal	National National National Regional Regional Regional Regional Regional	English English English English English English English French	2005–2015
Pillod [12]	The Globe and Mail	National	English	2008–2020
Brandt [13]	The Toronto Star The Calgary Herald The Vancouver Sun The Chronicle Herald The Winnipeg Free Press Times and Transcript	National Regional Regional Regional Regional Regional	English English English English English English	2015–2018
Hase et al. [14]	The Globe and Mail The Toronto Star	National National	English English	2006–2018
Sachdeva and McCaffrey [15]	n/a	n/a	English	1986–2016
Stoddart and Yang [16]	The Globe and Mail	National	English	2015–2020

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Study	Newspaper(s)	Scope	Language	Time Period
	The National Post	National	English	
	The Telegraph	Regional	English	
	The Telegram	Regional	English	
	The Chronicle Herald	Regional	English	
	The Guardian	Regional	English	
Chen et al. [17]	n/a	n/a	English	2018–2021

Note: This table summarizes the newspaper sources, geographic scope, language, and time periods covered by previous studies on climate change media coverage in Canada. Most studies focus exclusively on English-language national newspapers, particularly The Globe and Mail and The National Post.

Supplementary Table S2. Newspapers included in the CCF database.

Province/Territory	Newspaper	Language	Avg. Weekday Circ. (2015)	Articles
National	The Globe and Mail	English	226,171	29,442
	The National Post	English	79,157	20,032
	The Toronto Star	English	192,084	46,980
Alberta	The Edmonton Journal	English	59,244	18,162
	The Calgary Herald	English	58,865	19,336
British Columbia	The Vancouver Sun	English	78,901	17,871
	The Times Colonist	English	47,556	11,800
Manitoba	The Winnipeg Free Press	English	67,503	12,421
New Brunswick	L'Acadie Nouvelle	French	18,102	5,143
Newfoundland & Labrador	The Telegram	English	15,380	5,841
Nova Scotia	The Chronicle Herald	English	71,304	10,770
Ontario	The Toronto Sun	English	85,775	3,174
	Le Droit	French	28,616	4,727
Quebec	Le Devoir	French	28,729	13,685
	Le Journal de Montréal	French	175,220	5,458
	La Presse	French	98,657	6,917
	La Presse+ ^a	French	–	9,548
	The Montreal Gazette	English	53,344	9,567
Saskatchewan	The Star Phoenix	English	31,081	7,794
Yukon	The Whitehorse Daily Star	English	945	7,603
Total				266,271

^a In 2013, La Presse launched a free digital tablet edition called La Presse+ which offers slightly different content. The corpus considers articles published both on the website and in the digital tablet edition. Duplicates were discarded.

Supplementary Table S3. Complete CCF annotation framework.

#	Category	Code	Description
THEMATIC FRAMES			
Economic Frame			
1	Economic Frame (Primary Category)	economic_frame	If the sentence refers to climate change as an economic issue, including its negative or positive impacts on the economy, the financial costs or benefits of climate action, or the carbon footprint of the economic or industrial sector. This label captures any framing of climate change in economic terms, such as effects on jobs, markets, industries, or public and private finances.
2	Negative impacts of climate change on the economy	eco_neg_impact	If the sentence refers to the negative effects of climate change on the economy, such as damage to infrastructure, loss of agricultural productivity, disruptions to trade, increased costs for businesses, or economic decline in sectors vulnerable to climate shifts. This includes any economic losses directly resulting from changing climate conditions.
3	Positive impacts of climate change on the economy	eco_pos_impact	If the sentence refers to the positive effects of climate change on the economy, such as new agricultural opportunities, the opening of new trade routes, or the potential for economic growth in sectors adapting to climate shifts. This includes any economic gains directly resulting from changing climate conditions.
4	Economic disadvantages of climate action	eco_cost	If the sentence refers to the negative economic impacts caused by climate policies or climate action, such as costs associated with transitioning to green energy, investments in climate adaptation, debt incurred from financing climate action, or short-term job losses in fossil fuel industries. This category captures financial burdens and challenges linked to climate action or policy.
5	Economic benefits of climate action	eco_benefit	If the sentence refers to the positive economic impacts resulting from climate policies or climate action, such as job creation in renewable energy sectors, investments in green technologies, economic growth driven by sustainability initiatives, or cost savings from energy efficiency. This category captures financial gains and opportunities linked to climate action or policy.
6	Carbon footprint of the economic sector	eco_footprint	If the sentence refers to the environmental or carbon footprint of the economic or industrial sectors, including emissions and resource use related to manufacturing, transportation, energy production, mining, agriculture, and other commercial or industrial activities. This label covers the ecological impact generated by economic and industrial processes, not the effects of climate change on these sectors.
Health Frame			

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#	Category	Code	Description
7	Health Frame (Primary Category)	health_frame	If the sentence refers to climate change as a public health issue, including its negative or positive impacts on physical or mental health, the health co-benefits of climate action (e.g., cleaner air, active transportation), or the carbon footprint of the health sector. This label captures any framing of climate change in relation to health outcomes, healthcare systems, or health-related consequences and solutions.
8	Negative impacts of climate change on health	health_neg_impact	If the sentence refers to the negative effects of climate change on human health, such as increased risks of infectious diseases (e.g., malaria, dengue), respiratory and cardiovascular illnesses due to air pollution and heat exposure, heat-related illnesses (e.g., heatstroke), and mental health disorders (e.g., anxiety, depression) linked to environmental changes.
9	<i>Positive impacts of climate change on health*</i>	health_pos_impact	If the sentence refers to the potential positive effects of climate change on human health, such as reductions in cold-related illnesses and deaths, or increased opportunities for outdoor activity due to milder winters.
10	Health co-benefits of climate action	health_cobenefit	If the sentence refers to health benefits resulting from climate policies or actions, such as improved air quality, better nutrition, reduced respiratory or cardiovascular diseases, fewer premature deaths, and other positive health outcomes linked to climate mitigation and adaptation efforts.
11	<i>Carbon footprint of the health sector*</i>	health_footprint	If the sentence refers to the environmental or carbon footprint of the health sector, including emissions and resource use associated with healthcare facilities, medical equipment, pharmaceuticals production, patient and staff travel, and waste generated by medical activities. This label covers the ecological impact of delivering health services, not the impacts of climate change on health outcomes.
<i>Security Frame</i>			
12	Security Frame (Primary Category)	security_frame	If the sentence refers to the effects of climate change on security, such as violent conflict driven by resource scarcity, an influx of climate refugees, disruption of military operations, or the carbon footprint of the defense sector. This category includes broader security challenges arising from climate change, including geopolitical tensions or the strategic responses to climate impacts.
13	Presence of climate refugees	security_refugees	If the sentence refers to the displacement of populations or the presence of climate refugees as a result of a natural disaster, environmental degradation, or climate change impact. This category captures the movement of people fleeing climate-related disasters.

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#	Category	Code	Description
14	Conflict	security_conflict	If the sentence refers to the emergence or intensification of conflicts over natural resource exploitation due to climate change, such as disputes over water, land, or energy resources exacerbated by shifting climate patterns. This includes conflicts arising from scarcity or competition for increasingly stressed resources.
15	<i>Post-disaster military assistance*</i>	security_military	If the sentence refers to the deployment of military forces as reinforcement after a natural disaster, such as wildfires, floods, or other extreme events. This includes military support in disaster response and recovery, like providing humanitarian aid or maintaining order during a crisis.
16	<i>Disruption of military operations*</i>	security_disruption	If the sentence refers to the disruption of military operations due to a natural disaster affecting a military base or facilities. This includes cases where climate-induced events damage or disrupt military infrastructure, affecting defense readiness or response capabilities.
<i>Justice Frame</i>			
17	Justice Frame (Primary Category)	justice_frame	If the sentence frames climate change as a justice or moral issue, including references to who benefits or loses from climate action, differentiated responsibilities across countries or groups, unequal impacts of climate change, disparities in access to mitigation or adaptation measures, or concerns about intergenerational justice. This label captures ethical, equity-based, or fairness-oriented dimensions of climate discourse.
18	Winners and losers of climate action	justice_winners	If the sentence refers to how climate policies or actions create winners and losers, by benefiting certain groups (e.g., vulnerable populations, Indigenous communities, developing countries) while disadvantaging others (e.g., fossil fuel workers, carbon-intensive industries, or developed nations facing steeper transitions). This label focuses on distributional outcomes of climate policy.
19	Differentiated responsibility	justice_responsibility	If the sentence refers to the unequal responsibility for causing climate change, especially in reference to the principle of common but differentiated responsibilities. This includes statements emphasizing that developed countries, multinational corporations, or high-emitting individuals have historically contributed more to global warming and should therefore bear a greater share of the mitigation burden.
20	Unequal vulnerability to climate change	justice_vulnerability	If the sentence highlights how different populations are unequally affected by the impacts of climate change, due to geographic, social, or economic vulnerabilities. This includes references to marginalized groups such as women, seniors, racialized populations, people in developing countries, or those living in high-risk zones (e.g., coastal or arid regions). This label focuses on impacts, not responsibility or capacity.

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#	Category	Code	Description
21	Unequal access to climate action	justice_access	If the sentence refers to unequal capacity to act on climate change, emphasizing that countries, companies, or individuals do not have equal financial, technical, or institutional means to take climate action. For example, small businesses may struggle to decarbonize compared to multinational firms, and low-income households may not afford sustainable alternatives. This label reflects inequality in ability to act, not impacts or blame.
22	Intergenerational justice	justice_intergen	If the sentence refers to intergenerational justice, meaning that current decisions and actions should not compromise the rights or well-being of future generations. Do not apply this label to sentences that merely mention age differences or generational groups (e.g., “young people are protesting”) unless the statement clearly relates to moral responsibility across generations or the long-term consequences of climate inaction.
<i>Political Frame</i>			
23	Political Frame (Primary Category)	political_frame	If the sentence frames climate change as a political issue, including references to policy adoption or implementation, political debates or controversies, political positioning by parties or leaders on climate issues, and public opinion data related to climate change. This label captures the role of governance, decision-making, and political dynamics in climate discourse.
24	Policy action	pol_action	If the sentence refers to the formal approval, adoption, or enactment of a policy, plan, or strategy by a government at the local, provincial, national, or international level. This includes the passing of new legislation, regulations, government programs, or official commitments related to climate or environmental issues (e.g., approval of a carbon pricing law, ratification of a renewable energy policy, adoption of a climate action framework).
25	Political debate	pol_debate	If the sentence refers to a political debate or discussion about climate change, particularly focusing on the differing views or disagreements on climate policies, the effectiveness of past actions, or the responsibility of different levels of government (e.g., federal vs. provincial). This includes disagreements about specific climate initiatives, such as proposed or enacted policies, laws, or funding allocations.
26	Political positioning	pol_position	If the sentence refers to the explicit stance or position of a politician or political party on climate change or climate-related policies. This includes statements where a political figure or party expresses their support or opposition to specific climate actions (e.g., carbon taxes, energy transitions, or climate mitigation strategies), or when they make electoral promises regarding climate policies.

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#	Category	Code	Description
27	Public opinion data	pol_opinion	If the sentence refers to data, surveys, polls, or studies measuring public opinion, attitudes, beliefs, or perceptions related to climate change, environmental policies, or climate action. This label captures quantitative or qualitative information on how the public views climate issues.
<i>Scientific Frame</i>			
28	Scientific Frame (Primary Category)	scientific_frame	If the sentence frames climate change as a scientific issue, including references to scientific research, discoveries, explanations, or assessments of climate change. This includes mentions of scientific consensus or controversy, as well as expressions of uncertainty or certainty in climate science. This label captures the role of scientific knowledge and expertise in understanding and communicating climate change.
29	Scientific debate	sci_debate	If the sentence refers to a past or ongoing debate or disagreement within the scientific community about aspects of climate science or climate-related technologies. This includes contested interpretations of data, disputes about methodologies, or ethical debates surrounding proposed scientific or technological solutions such as geoengineering. It does not include general questioning of climate science or statements that challenge the validity or sufficiency of climate evidence.
30	Popularisation or scientific discovery	sci_discovery	If the sentence explains or describes climate science mostly grounded in the natural sciences—such as chemistry, physics, or biology—or presents scientific or technological discoveries related to climate change. This includes explanations of mechanisms (e.g., the greenhouse effect, climate feedbacks), climate modeling, or innovations such as carbon capture. It does not apply to statements that merely describe environmental conditions or trends without explaining an underlying scientific process.
31	Questioning of climate science	sci_skepticism	If the sentence questions the validity, accuracy, or sufficiency of climate science. This includes claims that climate projections are flawed, evidence is lacking, or conclusions are exaggerated. It only applies when science is being challenged or delegitimized, not when scientists acknowledge limitations, complexity, or areas of ongoing research.
32	Defense of climate science	sci_defense	If the sentence defends the robustness or credibility of climate science in response to skepticism or denial. This includes assertions that reinforce the scientific consensus, affirm the validity of climate models, or rebut claims that climate science is incorrect or misleading. It only applies when climate science is defended, not when it is neutrally presented.
<i>Environmental Frame</i>			

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
33	Environmental Frame (Primary Category)	environmental_ frame	If the sentence frames climate change as an environmental issue, including references to the degradation or disappearance of natural habitats, loss of biodiversity, and negative impacts on fauna and flora. This label captures ecological consequences of climate change and emphasizes the relationship between climate disruption and the natural environment.
34	Loss of natural environments	env_habitat	If the sentence refers to the long-term or irreversible disappearance or degradation of natural environments due to climate change. This includes situations where ecosystems or environmental features (e.g., glaciers, coral reefs) are predicted or expected to permanently disappear or significantly decline as a result of climate impacts, indicating that they will not recover. It does not apply to short-term, isolated, or recoverable events, such as one-time disasters (e.g., a single wildfire or a temporary drought) that might affect natural environments but do not necessarily imply permanent loss.
35	Loss of fauna and flora	env_species	If the sentence refers to the negative impacts of climate change on animal and plant species, including changes in habitat, shifts in migration or breeding patterns, increased mortality, loss of biodiversity, or threats to species survival. This label covers both temporary and ongoing effects on fauna and flora caused by climate change.
<i>Cultural Frame</i>			
36	Cultural Frame (Primary Category)	cultural_frame	If the sentence frames climate change as a cultural issue, including cultural depictions of climate change through art, literature, music, or film; challenges to hosting artistic or sports events due to climate impacts; the loss or disruption of Indigenous cultural practices; or the carbon footprint of the cultural and creative sectors. This label captures how climate change affects and is expressed through cultural life and identities.
37	Artistic representation	cult_art	If the sentence refers to cultural representations of climate change in various forms such as art, literature, music, film, documentaries, theater, or other creative expressions. This label captures how climate change is portrayed, interpreted, or communicated through cultural and artistic mediums.
38	Difficulty to host cultural or sports events	cult_event_ impact	If the sentence refers to how climate change disrupts or challenges the organization of artistic, cultural, or sports events, such as cancelled festivals, concerts, or competitions due to extreme weather (e.g., storms, heatwaves, wildfires, or lack of snow for winter sports). This label captures climate-related obstacles to event planning or continuity.
39	Loss of indigenous practices	cult_indigenous	If the sentence refers to the erosion or disruption of traditional cultural practices among Indigenous communities due to climate change, such as changes in animal migration, plant cycles, or land access that affect food systems, ceremonies, or knowledge transmission. This label emphasizes climate-driven impacts on Indigenous cultural continuity.

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
40	Carbon footprint of the cultural and sports sectors	cult_footprint	If the sentence refers to the environmental or carbon footprint of the cultural, artistic, or sports sectors, including the emissions and resource use associated with activities such as touring, international travel, energy-intensive venues, unsustainable stage or set production, or event-related waste. This label covers the ecological impact of producing and hosting cultural or sports events, not the impacts of climate on those events.
PRIMARY CATEGORIES			
<i>Actors/Messengers</i>			
41	Presence of Messengers (Primary Category)	messenger	If the sentence includes mentions or quotes of persons or organizations by journalists, whether through direct quotations or indirect references. This label captures the presence of sources, experts, officials, activists, or other actors cited or named within the text. A single individual or organization may be associated with multiple types of expertise or roles (e.g., a doctor would fall both under Health expert and Natural scientist).
42	Health expert	msg_health	If the quoted or reported statement comes from a person or organization with medical or health-related recognized expertise. This includes individuals working in the fields of medicine, public health, healthcare administration, or biomedical research, whether in a professional, academic, or governmental capacity (e.g., professor of public health, doctor, nurse, family physician, public health officer, WHO representative, Red Cross medical coordinator, Doctors without Borders spokesperson, Canadian Medical Association representative, certified health consultants, minister of Health).
43	Economic expert	msg_economic	If the quoted or reported statement comes from a person or organization with economic or financial recognized expertise. This includes individuals working in the fields of economics, finance, business, or economic policy, whether in a professional, academic, private sector, or governmental capacity (e.g., economist, finance minister, central bank official, business executive, investment analyst, professor of economics, representative of a chamber of commerce, World Bank or IMF spokesperson, economic think tank expert, labour market analyst).
44	Security expert	msg_security	If the quoted or reported statement comes from a person or organization with recognized expertise in security or defence matters. This includes individuals with academic, professional, or operational experience in military, national security, intelligence, or strategic affairs (e.g., professor in defense or security studies, active or retired member of the armed forces, intelligence agency representative, conflict resolution specialist, national defence minister, security analyst, representative of a security-focused think tank or NGO).

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
45	Legal expert	msg_legal	If the quoted or reported statement comes from a person or organization with recognized expertise in law or legal affairs. This includes individuals working in legal professions, legal academics, the judiciary, or justice-related government roles (e.g., lawyer, prosecutor, law professor, minister of justice, attorney general, legal aid representative, bar association representative, legal consultants).
46	Cultural or Sport expert	msg_cultural	If the quoted or reported statement comes from a person with recognized expertise in sports, arts, or culture (e.g., Olympian, coach, professor of literature, museum director, filmmaker, writer, festival organizer, musician, artist, indigenous knowledge keeper).
47	Natural scientist	msg_scientist	If the quoted or reported statement comes from a person or organization with recognized expertise in the natural or hard sciences. This includes individuals working in fields such as climatology, environmental science, physics, chemistry, biology, earth sciences, engineering, or related disciplines, whether in academic, governmental, or research institutions (e.g., climate scientist, environmental researcher, oceanographer, physicist, ecologist, geologist, engineer, professor of atmospheric sciences, IPCC contributor, Environment Canada scientist, research institute spokesperson).
48	Social scientist	msg_social	If the quoted or reported statement comes from a person with expertise in social sciences, helping to understand how society perceives, responds to, and governs issues like climate change (e.g., sociologist, political scientist, media scholar, economist).
49	Activist	msg_activist	If the quoted or reported statement comes from a person or organization known primarily for advocacy or activism. This includes individuals or groups engaged in campaigning, protesting, or raising awareness on environmental, social, or climate-related issues, whether through grassroots organizing, NGOs, or international movements (e.g., climate activist, environmental campaigner, Indigenous land defender, Fridays for Future organizer, Greenpeace spokesperson, Extinction Rebellion member, citizen activist, youth climate leader, environmental justice advocate).
50	Public official	msg_official	If the quoted or reported statement comes from a politician, elected official, government representative, or public decision-maker at any level of government. This includes individuals holding political office or involved in policymaking, regulation, or governance, whether municipal, provincial, federal, or international (e.g., prime minister, environment minister, member of parliament, city councillor, premier, senator, government spokesperson, cabinet member, UN climate negotiator, provincial regulator, parliamentary committee chair).
<i>Events</i>			

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#	Category	Code	Description
51	Presence of Events (Primary Category)	event	If the sentence refers to the occurrence or mention of one or more events. This includes extreme meteorological events (e.g., storms, floods, heatwaves), political or governmental events (e.g., elections, policy announcements, trials, judiciary decisions), meetings or conferences, publications, cultural events, protests, or other significant happenings.
52	Extreme meteorological event	evt_weather	If the sentence refers to the occurrence of an extreme meteorological or climate-related event that causes or risks causing significant disruption, damage, or harm. This includes events such as heatwaves, floods, wildfires, hurricanes, droughts, storms, or other weather-related disasters, whether sudden or prolonged.
53	Meeting	evt_meeting	If the sentence refers to an official visit, conference, summit, forum, meeting, convention, or assembly that brings together people to discuss, negotiate, or present on specific topics at the local, national, or international levels (e.g., town hall meeting, premiers' conference, ministerial roundtable, state visit, UN conference, international forum on clean energy, G7 meeting, COP28, World Economic Forum).
54	Publication	evt_publication	If the sentence refers to the release or publication of a document such as a report, academic article, survey results, op-ed, policy brief, or white paper, often aimed at informing the public, guiding decision-making, or contributing to public discourse (e.g., IPCC report, Statistics Canada survey, university study on climate impacts, editorial on climate policy, NGO report on emissions, think tank publication).
55	Election	evt_election	If the sentence refers to the occurrence of an election at the local, provincial, national, or international level, including general elections, by-elections, leadership races, referenda, or electoral campaigns (e.g., federal election, municipal by-election, provincial leadership race, EU parliamentary elections).
56	New policy	evt_policy	If the sentence refers to the announcement or unveiling of a policy, plan, or strategy at the local, provincial, national, or international level, including new legislation, regulations, government programs, or official commitments, including those related to climate or environmental issues (e.g., carbon pricing plan, renewable energy strategy, climate action framework).
57	Judiciary decision	evt_judiciary	If the sentence refers to the occurrence of a trial, court ruling, legal proceeding, or judicial decision at any level of the judicial system, including constitutional challenges, environmental lawsuits, regulatory hearings, or Supreme Court decisions, including those related to climate or environmental issues (e.g., court ruling on carbon pricing, environmental class action lawsuit, constitutional challenge to climate legislation, tribunal decision on pipeline approval).

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#	Category	Code	Description
58	Cultural or Sports event	evt_cultural	If the sentence refers to the organization or hosting of a sports, artistic, or cultural event (e.g., Olympics games, national hockey tournament, local marathon, film screening, music concert, theatre performance, book launch, mural festival, photography exhibit).
59	Protest	evt_protest	If the sentence refers to the organization of a protest (e.g., climate strike, anti-pipeline protest, extinction rebellion action, anti-racism march, women’s right rally, union-led protest for better wages).
<i>Solutions</i>			
60	Presence of Solutions (Primary Category)	solution	If the sentence refers to solutions addressing climate change, including mitigation efforts to reduce greenhouse gas emissions or adaptation strategies to manage climate impacts. This includes direct mentions of specific policies, technologies, behaviors, or initiatives, as well as indirect references to approaches aimed at combating or coping with climate change.
61	Mitigation strategy	sol_mitigation	If the sentence refers to mitigation solutions designed to reduce or prevent the emission of greenhouse gases and limit the extent of future climate change. These strategies aim to address the root causes of climate change by lowering carbon footprints, enhancing carbon sinks, or transitioning to low-carbon technologies (e.g., investing in renewable energy, improving energy efficiency, implementing carbon pricing, promoting public transit, reforestation, phasing out fossil fuels).
62	Adaptation strategy	sol_adaptation	If the sentence refers to adaptation solutions designed to reduce vulnerability to the negative effects of climate change by adjusting or preparing for those impacts. These strategies involve actions to manage the risks and enhance resilience, particularly by reducing exposure to climate-related hazards and minimizing harm to people, infrastructure, and ecosystems (e.g., providing cooling centres during heat waves, restoring wetlands to limit flooding, altering farming techniques to cope with changing weather patterns).
EMOTIONAL TONE			
63	Emotion: positive	tone_positive	If the tone is generally optimistic, reassuring, or enthusiastic, highlighting favorable aspects or expressing confidence in climate solutions or actions.
64	Emotion: negative	tone_negative	If the tone is generally alarming, critical, or pessimistic, especially if the sentence expresses concern, fear, or frustration about climate change or its impacts.
65	Emotion: neutral	tone_neutral	If the sentence is mainly informative or descriptive without an evaluative tone, presenting facts or balanced perspectives without strong emotional content.
GEOGRAPHIC FOCUS			

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#	Category	Code	Description
66	Mention of Canada	canada	If the sentence refers to Canada in any context, domestic or international, including but not limited to climate change impacts on Canadian territory, as well as Canada’s policies, actions, or roles.
URGENCY/ALARMISM			
67	Urgency to act	urgency	If the sentence conveys a strong sense of urgency or alarmism, emphasizing the immediate need for action or highlighting critical risks and dangers related to climate change, often with a tone that signals warning or crisis.

* Categories marked with an asterisk were included in the annotation protocol but had insufficient positive examples for model training. Health subcategories (`health_pos_impact`, `health_footprint`) were completely excluded due to insufficient annotations in both English and French. Security subcategories (`security_military`, `security_disruption`) are partially available (trained in some languages only). The framework includes 67 categories in total: 63 fully available, 2 partially available, and 2 excluded.

Supplementary Table S4. Complete model training performance metrics for all annotation categories.

#	Category	Code	F1 (Class 1)		F1 (Class 0)		Macro F1	
			EN	FR	EN	FR	EN	FR
THEMATIC FRAMES								
Economic Frame								
1	Economic Frame (Primary Category)	economic_frame	0.745	0.814	0.944	0.957	0.845	0.885
2	Negative impacts of climate change on the economy	eco_neg_impact	0.727	0.889	0.914	0.944	0.821	0.917
3	Positive impacts of climate change on the economy	eco_pos_impact	0.333	0.400	0.995	0.996	0.664	0.698
4	Economic disadvantages of climate action	eco_cost	0.615	0.500	0.848	0.913	0.732	0.707
5	Economic benefits of climate action	eco_benefit	0.500	0.516	0.952	0.981	0.726	0.749
6	Carbon footprint of the economic sector	eco_footprint	0.857	0.857	0.938	0.950	0.897	0.904
Health Frame								
7	Health Frame (Primary Category)	health_frame	0.800	0.667	0.989	0.994	0.894	0.830
8	Negative impacts of climate change on health	health_neg_impact	0.909	0.857	0.000	0.000	0.455	0.429
9	Health co-benefits of climate action	health_cobenefit	0.400	0.011	0.571	0.227	0.486	0.119
Security Frame								
10	Security Frame (Primary Category)	security_frame	0.870	0.800	0.996	0.997	0.933	0.898
11	Presence of climate refugees	security_refugees	1.000	1.000	1.000	1.000	1.000	1.000

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Supplementary Table S4 – Continued from previous page

#	Category	Code	F1 (Class 1)		F1 (Class 0)		Macro F1	
			EN	FR	EN	FR	EN	FR
12	Conflict	security_conflict	1.000	1.000	1.000	1.000	1.000	1.000
13	Post-disaster military assistance	security_military	–	0.010	–	0.043	–	0.026
14	Disruption of military operations	security_disruption	–	1.000	–	1.000	–	1.000
<i>Justice Frame</i>								
15	Justice Frame (Primary Category)	justice_frame	0.719	0.717	0.975	0.981	0.847	0.849
16	Winners and losers of climate action	justice_winners	0.667	0.667	0.933	0.923	0.800	0.795
17	Differentiated responsibility	justice_responsibility	0.857	0.750	0.909	0.750	0.883	0.750
18	Unequal vulnerability to climate change	justice_vulnerability	0.571	0.600	0.992	0.995	0.782	0.798
19	Unequal access to climate action	justice_access	0.364	0.625	0.000	0.993	0.182	0.809
20	Intergenerational justice	justice_intergen	0.933	0.800	0.999	0.999	0.966	0.899
<i>Political Frame</i>								
21	Political Frame (Primary Category)	political_frame	0.808	0.774	0.897	0.888	0.853	0.831
22	Policy action	pol_action	0.621	0.667	0.985	0.966	0.803	0.816
23	Political debate	pol_debate	0.916	0.966	0.222	0.571	0.569	0.768
24	Political positioning	pol_position	0.750	0.800	0.946	0.989	0.848	0.895
25	Public opinion data	pol_opinion	0.909	1.000	0.999	1.000	0.954	1.000
<i>Scientific Frame</i>								
26	Scientific Frame (Primary Category)	scientific_frame	0.784	0.702	0.953	0.962	0.869	0.832
27	Scientific debate	sci_debate	0.737	0.500	0.848	0.875	0.793	0.688

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Supplementary Table S4 – Continued from previous page

#	Category	Code	F1 (Class 1)		F1 (Class 0)		Macro F1	
			EN	FR	EN	FR	EN	FR
28	Popularisation or scientific discovery	sci_discovery	0.828	0.938	0.800	0.750	0.814	0.844
29	Questioning of climate science	sci_skepticism	0.727	0.333	0.927	0.995	0.827	0.664
30	Defense of climate science	sci_defense	0.571	0.000	0.933	0.919	0.752	0.459
<i>Environmental Frame</i>								
31	Environmental Frame (Primary Category)	environmental_frame	0.842	0.625	0.989	0.980	0.915	0.802
32	Loss of natural environments	env_habitat	1.000	0.857	1.000	0.800	1.000	0.829
33	Loss of fauna and flora	env_species	0.889	0.857	0.857	0.800	0.873	0.829
<i>Cultural Frame</i>								
34	Cultural Frame (Primary Category)	cultural_frame	0.773	0.833	0.986	0.993	0.879	0.913
35	Artistic representation	cult_art	1.000	1.000	1.000	1.000	1.000	1.000
36	Difficulty to host cultural or sports events	cult_event_impact	0.706	1.000	0.993	1.000	0.850	1.000
37	Loss of indigenous practices	cult_indigenous	1.000	0.026	1.000	0.899	1.000	0.462
38	Carbon footprint of the cultural and sports sectors	cult_footprint	1.000	0.005	1.000	0.143	1.000	0.074
PRIMARY CATEGORIES								
<i>Actors/Messengers</i>								
39	Presence of Messengers (Primary Category)	messenger	0.912	0.929	0.904	0.915	0.908	0.922
40	Health expert	msg_health	0.857	0.909	0.997	0.999	0.927	0.954
41	Economic expert	msg_economic	0.600	0.750	0.970	0.973	0.785	0.861
42	Security expert	msg_security	0.667	1.000	0.993	1.000	0.830	1.000

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Supplementary Table S4 – Continued from previous page

#	Category	Code	F1 (Class 1)		F1 (Class 0)		Macro F1	
			EN	FR	EN	FR	EN	FR
43	Legal expert	msg_legal	0.667	0.769	0.999	0.996	0.833	0.883
44	Cultural or Sport expert	msg_cultural	1.000	0.556	1.000	0.990	1.000	0.773
45	Natural scientist	msg_scientist	0.789	0.833	0.977	0.971	0.883	0.902
46	Social scientist	msg_social	0.769	0.645	0.996	0.986	0.883	0.816
47	Activist	msg_activist	0.571	1.000	0.978	1.000	0.775	1.000
48	Public official	msg_official	0.703	0.923	0.897	0.964	0.800	0.943
<i>Events</i>								
49	Presence of Events (Primary Category)	event	0.794	0.819	0.935	0.932	0.865	0.876
50	Extreme meteorological event	evt_weather	1.000	0.857	1.000	0.968	1.000	0.912
51	Meeting	evt_meeting	0.824	0.957	0.936	0.982	0.880	0.969
52	Publication	evt_publication	0.873	1.000	0.990	1.000	0.931	1.000
53	Election	evt_election	1.000	0.800	1.000	0.998	1.000	0.899
54	New policy	evt_policy	0.769	0.727	0.943	0.954	0.856	0.841
55	Judiciary decision	evt_judiciary	1.000	1.000	1.000	1.000	1.000	1.000
56	Cultural or Sports event	evt_cultural	0.333	1.000	0.995	1.000	0.664	1.000
57	Protest	evt_protest	0.889	0.727	0.999	0.996	0.944	0.862
<i>Solutions</i>								
58	Presence of Solutions (Primary Category)	solution	0.737	0.878	0.914	0.944	0.825	0.911
59	Mitigation strategy	sol_mitigation	0.750	0.812	0.935	0.942	0.842	0.877
60	Adaptation strategy	sol_adaptation	0.696	0.800	0.991	0.989	0.843	0.894
EMOTIONAL TONE								
61	Emotion: positive	tone_positive	0.526	0.690	0.966	0.968	0.746	0.829
62	Emotion: negative	tone_negative	0.706	0.765	0.785	0.843	0.745	0.804
63	Emotion: neutral	tone_neutral	0.741	0.789	0.676	0.693	0.709	0.741
GEOGRAPHIC FOCUS								

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Supplementary Table S4 – Continued from previous page

#	Category	Code	F1 (Class 1)		F1 (Class 0)		Macro F1	
			EN	FR	EN	FR	EN	FR
64	Mention of Canada	canada	0.942	0.964	0.968	0.980	0.955	0.972
URGENCY TO ACT								
65	Urgency to act	urgency	0.591	0.649	0.975	0.984	0.783	0.816
AVERAGE PERFORMANCE METRICS								
		English Average	0.769	–	0.905	–	0.837	–
		French Average	–	0.737	–	0.894	–	0.816
		Overall Average	0.753		0.900		0.826	
TOTAL: 65 ANNOTATION CATEGORIES								
(65 categories with at least one model; 0 categories entirely excluded*)								

Note. F1 scores at the best epoch retained by the combined-criterion best-model selector ($0.7 \cdot F_1^{(1)} + 0.3 \cdot \text{macro-}F_1$). EN columns report metrics for the English BERT-base-uncased classifiers; FR columns report metrics for the French CamemBERT-base classifiers. Categories with “–” in both columns could not be trained because the manual annotation set contained too few positive examples in either language. The dataset distribution behind each row is reported in Supplementary Table S5.

Supplementary Table S5. Training and validation dataset distribution across all annotation categories.

#	Category	Code	English				French			
			Training		Validation		Training		Validation	
			Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
THEMATIC FRAMES										
Economic Frame										
1	Economic Frame (Primary Category)	economic_frame	218	1067	24	118	252	1164	28	129
2	Negative impacts of climate change on the economy	eco_neg_impact	61	158	6	17	75	178	8	19
3	Positive impacts of climate change on the economy	eco_pos_impact	14	205	1	22	7	245	1	27
4	Economic disadvantages of climate action	eco_cost	63	156	6	17	42	211	4	23
5	Economic benefits of climate action	eco_benefit	36	183	3	20	48	205	5	22
6	Carbon footprint of the economic sector	eco_footprint	67	152	7	16	77	176	8	19
Health Frame										
7	Health Frame (Primary Category)	health_frame	57	1228	6	136	37	1379	4	153
8	Negative impacts of climate change on health	health_neg_impact	47	10	5	1	32	5	3	1
9	Health co-benefits of climate action	health_cobenefit	8	49	1	5	4	33	1	3
Security Frame										
10	Security Frame (Primary Category)	security_frame	19	1266	2	140	19	1397	2	155

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Supplementary Table S5 – Continued from previous page

#	Category	Code	English				French			
			Training		Validation		Training		Validation	
			Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
11	Presence of climate refugees	security_refugees	10	9	1	1	7	12	1	1
12	Conflict	security_conflict	6	13	1	1	7	12	1	1
13	Post-disaster military assistance	security_military	0	18	1	2	4	15	1	1
14	Disruption of military operations	security_disruption	0	18	1	2	1	18	1	1
<i>Justice Frame</i>										
15	Justice Frame (Primary Category)	justice_frame	90	1196	9	132	81	1335	9	148
16	Winners and losers of climate action	justice_winners	20	70	2	7	20	62	2	6
17	Differentiated responsibility	justice_responsibility	38	52	4	5	41	41	4	4
18	Unequal vulnerability to climate change	justice_vulnerability	8	81	1	9	9	72	1	8
19	Unequal access to climate action	justice_access	21	69	2	7	20	62	2	6
20	Intergenerational justice	justice_intergen	14	76	1	8	6	75	1	8
<i>Political Frame</i>										
21	Political Frame (Primary Category)	political_frame	416	869	46	96	440	977	48	108
22	Policy action	pol_action	62	355	6	39	58	382	6	42
23	Political debate	pol_debate	344	72	38	8	396	45	43	4
24	Political positioning	pol_position	61	356	6	39	35	405	3	45
25	Public opinion data	pol_opinion	21	396	2	43	20	420	2	46
<i>Scientific Frame</i>										

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Supplementary Table S5 – Continued from previous page

#	Category	Code	English				French			
			Training		Validation		Training		Validation	
			Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
26	Scientific Frame (Primary Category)	scientific_frame	243	1042	27	115	185	1232	20	136
27	Scientific debate	sci_debate	98	146	10	16	46	139	5	15
28	Popularisation or scientific discovery	sci_discovery	126	117	14	13	144	41	16	4
29	Questioning of climate science	sci_skepticism	50	194	5	21	17	169	1	18
30	Defense of climate science	sci_defense	37	207	4	22	32	153	3	17
<i>Environmental Frame</i>										
31	Environmental Frame (Primary Category)	environmental_frame	84	1201	9	133	63	1354	6	150
32	Loss of natural environments	env_habitat	50	35	5	3	35	28	3	3
33	Loss of fauna and flora	env_species	42	43	4	4	32	31	3	3
<i>Cultural Frame</i>										
34	Cultural Frame (Primary Category)	cultural_frame	44	1242	4	137	45	1371	5	152
35	Artistic representation	cult_art	20	24	2	2	18	27	2	3
36	Difficulty to host cultural or sports events	cult_event_impact	11	33	1	3	21	25	2	2
37	Loss of indigenous practices	cult_indigenous	10	34	1	3	4	41	1	4
38	Carbon footprint of the cultural and sports sectors	cult_footprint	1	42	1	4	2	43	1	4
PRIMARY CATEGORIES										
<i>Actors/Messengers</i>										

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Supplementary Table S5 – Continued from previous page

#	Category	Code	English				French			
			Training		Validation		Training		Validation	
			Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
39	Presence of Messengers (Primary Category)	messenger	657	629	72	69	736	681	81	75
40	Health expert	msg_health	10	647	1	71	8	728	1	80
41	Economic expert	msg_economic	55	602	6	66	67	669	7	74
42	Security expert	msg_security	3	653	1	72	10	726	1	80
43	Legal expert	msg_legal	1	655	1	72	4	731	1	81
44	Cultural or Sport expert	msg_cultural	13	644	1	71	9	727	1	80
45	Natural scientist	msg_scientist	107	550	11	61	117	620	12	68
46	Social scientist	msg_social	9	648	1	71	26	711	2	78
47	Activist	msg_activist	53	604	5	67	55	681	6	75
48	Public official	msg_official	171	486	18	54	223	513	24	57
<i>Events</i>										
49	Presence of Events (Primary Category)	event	298	987	33	109	355	1062	39	117
50	Extreme meteorological event	evt_weather	72	226	8	25	59	297	6	32
51	Meeting	evt_meeting	75	224	8	24	100	255	11	28
52	Publication	evt_publication	74	225	8	24	114	242	12	26
53	Election	evt_election	16	283	1	31	18	337	2	37
54	New policy	evt_policy	55	243	6	27	60	296	6	32
55	Judiciary decision	evt_judiciary	10	288	1	32	10	345	1	38
56	Cultural or Sports event	evt_cultural	2	296	1	32	11	344	1	38
57	Protest	evt_protest	11	288	1	31	8	347	1	38
<i>Solutions</i>										
58	Presence of Solutions (Primary Category)	solution	314	972	34	107	415	1001	46	111
59	Mitigation strategy	sol_mitigation	279	35	31	3	360	55	40	6

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Supplementary Table S5 – Continued from previous page

#	Category	Code	English				French			
			Training		Validation		Training		Validation	
			Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
60	Adaptation strategy	sol_adaptation	46	268	5	29	64	351	7	39
EMOTIONAL TONE										
61	Emotion: positive	tone_positive	103	1182	11	131	151	1266	16	140
62	Emotion: negative	tone_negative	491	794	54	88	537	880	59	97
63	Emotion: neutral	tone_neutral	686	599	76	66	726	691	80	76
GEOGRAPHIC FOCUS										
64	Mention of Canada	canada	445	840	49	93	493	924	54	102
URGENCY TO ACT										
65	Urgency to act	urgency	66	1219	7	135	63	1354	6	150

Note. Positive (**Pos**) and negative (**Neg**) sentence counts in the training and validation splits, by language. Categories with zero counts in every split could not be trained due to insufficient positive examples; the resulting per-category training metrics are reported in Supplementary Table S4.

Supplementary Table S6. Named Entity Recognition performance.

Table S11: Named Entity Recognition (**ner_entities** variable) model performance metrics

Language	Model	PER F1	ORG F1	LOC F1
English	BERT-base-NER	0.961	0.811	0.925
French	spaCy fr_core_news_lg	0.880	–	–
	CamemBERT-NER	–	0.824	0.929

Note: The hybrid approach for French combines spaCy for person entities (PER) with CamemBERT-NER for organization (ORG) and location (LOC) entities based on empirical evaluation on the dataset.

Performance metrics are from the original model documentation.

Supplementary Table S7. Detailed validation performance metrics on the 1,000-sentence gold standard.

#	Category	Code	F1 Macro			F1 Micro			F1 Weighted			Support		
			EN	FR	ALL	EN	FR	ALL	EN	FR	ALL	EN	FR	ALL
THEMATIC FRAMES														
Economic Frame														
1	Economic Frame (Primary Category)	economic_frame	0.847	0.814	0.830	0.892	0.864	0.878	0.890	0.862	0.875	120	129	249
2	Negative impacts of climate change on the economy	eco_neg_impact	0.820	0.741	0.783	0.848	0.810	0.828	0.856	0.833	0.842	26	18	44
3	Positive impacts of climate change on the economy	eco_pos_impact	0.806	0.508	0.684	0.905	0.845	0.873	0.913	0.908	0.900	12	1	13
4	Economic disadvantages of climate action	eco_cost	0.741	0.702	0.723	0.800	0.802	0.801	0.813	0.806	0.810	22	23	45
5	Economic benefits of climate action	eco_benefit	0.885	0.817	0.848	0.924	0.871	0.896	0.928	0.872	0.899	19	26	45
6	Carbon footprint of the economic sector	eco_footprint	0.785	0.788	0.787	0.838	0.845	0.842	0.831	0.843	0.837	30	29	59
Health Frame														
7	Health Frame (Primary Category)	health_frame	0.800	0.817	0.808	0.947	0.961	0.954	0.947	0.964	0.956	35	23	58
8	Negative impacts of climate change on health	health_neg_impact	0.386	0.340	0.364	0.629	0.514	0.571	0.485	0.349	0.416	22	18	40
9	Health co-benefits of climate action	health_cobenefit	0.252	0.103	0.186	0.257	0.114	0.186	0.301	0.023	0.177	4	4	8
Security Frame														
10	Security Frame (Primary Category)	security_frame	0.825	0.730	0.782	0.951	0.939	0.945	0.955	0.950	0.952	30	19	49
11	Presence of climate refugees	security_refugees	0.476	0.417	0.481	0.545	0.429	0.488	0.468	0.476	0.459	21	6	27
12	Conflict	security_conflict	0.358	0.286	0.324	0.364	0.286	0.326	0.398	0.286	0.346	7	6	13
13	Post-disaster military assistance	security_military	0.000	0.045	0.045	0.000	0.048	0.048	0.000	0.004	0.004	0	2	2
14	Disruption of military operations	security_disruption	0.000	1.000	1.000	0.000	1.000	1.000	0.000	1.000	1.000	0	2	2
Justice Frame														
15	Justice Frame (Primary Category)	justice_frame	0.832	0.880	0.857	0.917	0.937	0.927	0.921	0.939	0.930	62	74	136

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#	Category	Code	F1 Macro			F1 Micro			F1 Weighted			Support		
			EN	FR	ALL	EN	FR	ALL	EN	FR	ALL	EN	FR	ALL
16	Winners and losers of climate action	justice_winners	0.654	0.737	0.708	0.765	0.762	0.764	0.802	0.775	0.786	10	22	32
17	Differentiated responsibility	justice_responsibility	0.900	0.823	0.860	0.926	0.833	0.879	0.927	0.838	0.883	19	27	46
18	Unequal vulnerability to climate change	justice_vulnerability	0.786	0.894	0.845	0.852	0.917	0.885	0.858	0.918	0.888	16	21	37
19	Unequal access to climate action	justice_access	0.475	0.796	0.651	0.481	0.833	0.661	0.456	0.835	0.674	27	23	50
20	Intergenerational justice	justice_intergen	0.924	0.952	0.942	0.963	0.964	0.964	0.965	0.965	0.965	10	19	29
<i>Political Frame</i>														
21	Political Frame (Primary Category)	political_frame	0.812	0.842	0.829	0.827	0.844	0.836	0.829	0.845	0.837	167	217	384
22	Policy action	pol_action	0.802	0.722	0.756	0.914	0.864	0.886	0.909	0.854	0.879	26	38	64
23	Political debate	pol_debate	0.469	0.650	0.564	0.694	0.820	0.763	0.595	0.781	0.699	127	175	302
24	Political positioning	pol_position	0.752	0.808	0.782	0.801	0.912	0.862	0.821	0.920	0.877	35	24	59
25	Public opinion data	pol_opinion	0.877	0.910	0.894	0.962	0.974	0.969	0.966	0.976	0.971	12	15	27
<i>Scientific Frame</i>														
26	Scientific Frame (Primary Category)	scientific_frame	0.891	0.890	0.890	0.951	0.949	0.950	0.948	0.947	0.947	74	75	149
27	Scientific debate	sci_debate	0.962	0.867	0.911	0.962	0.869	0.912	0.962	0.869	0.912	26	26	52
28	Popularisation or scientific discovery	sci_discovery	0.884	0.797	0.838	0.885	0.803	0.841	0.886	0.803	0.842	31	36	67
29	Questioning of climate science	sci_skepticism	0.783	0.768	0.775	0.846	0.836	0.841	0.860	0.857	0.858	9	9	18
30	Defense of climate science	sci_defense	0.769	0.460	0.682	0.846	0.852	0.850	0.862	0.785	0.843	8	9	17
<i>Environmental Frame</i>														
31	Environmental Frame (Primary Category)	environmental_frame	0.866	0.875	0.871	0.967	0.967	0.967	0.969	0.968	0.969	28	32	60
32	Loss of natural environments	env_habitat	0.804	0.573	0.721	0.806	0.707	0.753	0.802	0.637	0.730	16	26	42
33	Loss of fauna and flora	env_species	0.714	0.692	0.703	0.722	0.707	0.714	0.717	0.694	0.705	19	21	40
<i>Cultural Frame</i>														
34	Cultural Frame (Primary Category)	cultural_frame	0.802	0.581	0.708	0.923	0.880	0.901	0.931	0.920	0.921	41	11	52
35	Artistic representation	cult_art	0.834	0.469	0.659	0.851	0.559	0.704	0.858	0.649	0.739	18	6	24
36	Difficulty to host cultural or sports events	cult_event_impact	0.631	0.354	0.484	0.731	0.456	0.593	0.777	0.595	0.686	8	2	10
37	Loss of indigenous practices	cult_indigenous	0.824	0.415	0.595	0.910	0.618	0.763	0.922	0.749	0.825	7	1	8

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Supplementary Table S7 – Continued from previous page

#	Category	Code	F1 Macro			F1 Micro			F1 Weighted			Support		
			EN	FR	ALL	EN	FR	ALL	EN	FR	ALL	EN	FR	ALL
38	Carbon footprint of the cultural and sports sectors	cult_footprint	0.702	0.014	0.351	0.925	0.015	0.467	0.945	0.000	0.613	2	1	3
PRIMARY CATEGORIES														
<i>Actors/Messengers</i>														
39	Presence of Messengers (Primary Category)	messenger	0.955	0.967	0.961	0.957	0.969	0.963	0.957	0.969	0.963	301	312	613
40	Health expert	msg_health	0.862	0.951	0.907	0.964	0.987	0.975	0.966	0.987	0.976	19	21	40
41	Economic expert	msg_economic	0.847	0.895	0.874	0.954	0.961	0.957	0.949	0.963	0.957	30	28	58
42	Security expert	msg_security	0.728	0.813	0.787	0.970	0.958	0.964	0.973	0.962	0.968	7	14	21
43	Legal expert	msg_legal	0.861	0.927	0.914	0.990	0.980	0.985	0.989	0.982	0.986	7	19	26
44	Cultural or Sport expert	msg_cultural	0.853	0.825	0.843	0.954	0.967	0.961	0.954	0.969	0.961	26	13	39
45	Natural scientist	msg_scientist	0.920	0.896	0.907	0.960	0.941	0.951	0.960	0.944	0.952	44	46	90
46	Social scientist	msg_social	0.938	0.948	0.944	0.987	0.984	0.985	0.987	0.984	0.986	17	24	41
47	Activist	msg_activist	0.934	0.903	0.918	0.977	0.964	0.970	0.976	0.963	0.970	31	33	64
48	Public official	msg_official	0.927	0.930	0.929	0.934	0.935	0.934	0.934	0.935	0.934	104	113	217
<i>Events</i>														
49	Presence of Events (Primary Category)	event	0.881	0.912	0.897	0.886	0.913	0.900	0.885	0.913	0.900	201	222	423
50	Extreme meteorological event	evt_weather	0.887	0.891	0.889	0.935	0.955	0.946	0.938	0.958	0.949	29	21	50
51	Meeting	evt_meeting	0.855	0.891	0.877	0.908	0.914	0.911	0.914	0.918	0.916	30	50	80
52	Publication	evt_publication	0.950	0.962	0.958	0.973	0.968	0.970	0.973	0.969	0.971	30	62	92
53	Election	evt_election	0.911	0.895	0.903	0.962	0.964	0.963	0.960	0.963	0.962	25	22	47
54	New policy	evt_policy	0.821	0.837	0.829	0.886	0.914	0.901	0.895	0.915	0.906	29	33	62
55	Judiciary decision	evt_judiciary	0.628	0.939	0.808	0.903	0.982	0.946	0.927	0.981	0.951	6	19	25
56	Cultural or Sports event	evt_cultural	0.906	0.857	0.881	0.973	0.964	0.968	0.971	0.967	0.968	17	12	29
57	Protest	evt_protest	0.982	0.909	0.943	0.995	0.973	0.983	0.995	0.973	0.983	16	18	34
<i>Solutions</i>														
58	Presence of Solutions (Primary Category)	solution	0.891	0.909	0.900	0.921	0.931	0.926	0.922	0.930	0.926	114	133	247
59	Mitigation strategy	sol_mitigation	0.872	0.801	0.839	0.884	0.839	0.861	0.882	0.850	0.864	76	97	173
60	Adaptation strategy	sol_adaptation	0.889	0.903	0.895	0.934	0.952	0.943	0.934	0.954	0.944	22	16	38
EMOTIONAL TONE														
61	Emotion: positive	tone_positive	0.712	0.692	0.702	0.902	0.892	0.897	0.910	0.904	0.907	38	37	75
62	Emotion: negative	tone_negative	0.793	0.804	0.799	0.807	0.813	0.810	0.814	0.819	0.816	151	165	316
63	Emotion: neutral	tone_neutral	0.756	0.798	0.777	0.764	0.809	0.787	0.765	0.807	0.787	295	306	601

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#	Category	Code	F1 Macro			F1 Micro			F1 Weighted			Support		
			EN	FR	ALL	EN	FR	ALL	EN	FR	ALL	EN	FR	ALL
GEOGRAPHIC FOCUS														
64	Mention of Canada	canada	0.986	0.978	0.982	0.986	0.978	0.982	0.986	0.978	0.982	235	255	490
URGENCY TO ACT														
65	Urgency to act	urgency	0.874	0.835	0.853	0.976	0.965	0.970	0.977	0.967	0.972	23	25	48
OVERALL PERFORMANCE		All Categories	0.869	0.864	0.866	0.909	0.902	0.905	0.911	0.905	0.908	3069	3332	6401

Note. F1 macro, F1 micro, F1 weighted, and positive-class support computed on the 1,000-sentence gold standard. EN, FR, and ALL columns report metrics on the English subset, the French subset, and their union. **OVERALL PERFORMANCE** in the last row reports the corpus-wide aggregates rather than the mean of the per-category scores. Inter-coder agreement statistics on the same 1,000 sentences are reported in Supplementary Tables S9 and S10.

Supplementary Table S8. Database-wide distribution of annotation categories across the 9.2 million sentences of CCF_processed_data.

#	Category	Code	Count			Proportion (%)		
			EN	FR	ALL	EN	FR	ALL
THEMATIC FRAMES								
Economic Frame								
1	Economic Frame (Primary Category)	economic_frame	1,232,910	185,032	1,417,942	15.94	12.64	15.41
2	Negative impacts of climate change on the economy	eco_neg_impact	193,068	30,670	223,738	2.50	2.10	2.43
3	Positive impacts of climate change on the economy	eco_pos_impact	1,153	1,949	3,102	0.01	0.13	0.03
4	Economic disadvantages of climate action	eco_cost	516,374	40,259	556,633	6.68	2.75	6.05
5	Economic benefits of climate action	eco_benefit	75,145	30,266	105,411	0.97	2.07	1.15
6	Carbon footprint of the economic sector	eco_footprint	292,909	61,974	354,883	3.79	4.23	3.86
Health Frame								
7	Health Frame (Primary Category)	health_frame	106,679	21,164	127,843	1.38	1.45	1.39
8	Negative impacts of climate change on health	health_neg_impact	106,679	21,164	127,843	1.38	1.45	1.39
9	Health co-benefits of climate action	health_cobenefit	63,774	19,595	83,369	0.82	1.34	0.91
Security Frame								
10	Security Frame (Primary Category)	security_frame	32,925	10,255	43,180	0.43	0.70	0.47

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#	Category	Code	Count			Proportion (%)		
			EN	FR	ALL	EN	FR	ALL
11	Presence of climate refugees	security_refugees	21,384	4,248	25,632	0.28	0.29	0.28
12	Conflict	security_conflict	20,405	9,015	29,420	0.26	0.62	0.32
13	Post-disaster military assistance	security_military	0	10,253	10,253	0.00	0.70	0.11
14	Disruption of military operations	security_disruption	0	4	4	0.00	0.00	0.00
<i>Justice Frame</i>								
15	Justice Frame (Primary Category)	justice_frame	276,401	20,938	297,339	3.57	1.43	3.23
16	Winners and losers of climate action	justice_winners	45,550	8,905	54,455	0.59	0.61	0.59
17	Differentiated responsibility	justice_responsibility	59,556	9,084	68,640	0.77	0.62	0.75
18	Unequal vulnerability to climate change	justice_vulnerability	13,194	2,188	15,382	0.17	0.15	0.17
19	Unequal access to climate action	justice_access	215,212	4,791	220,003	2.78	0.33	2.39
20	Intergenerational justice	justice_intergen	26,797	1,160	27,957	0.35	0.08	0.30
<i>Political Frame</i>								
21	Political Frame (Primary Category)	political_frame	2,408,280	409,831	2,818,111	31.13	28.00	30.64
22	Policy action	pol_action	204,684	18,833	223,517	2.65	1.29	2.43
23	Political debate	pol_debate	2,357,345	401,652	2,758,997	30.47	27.44	29.99
24	Political positioning	pol_position	860,176	49,748	909,924	11.12	3.40	9.89
25	Public opinion data	pol_opinion	46,825	8,186	55,011	0.61	0.56	0.60
<i>Scientific Frame</i>								

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Supplementary Table S8 – Continued from previous page

#	Category	Code	Count			Proportion (%)		
			EN	FR	ALL	EN	FR	ALL
26	Scientific Frame (Primary Category)	scientific_ frame	516,670	75,144	591,814	6.68	5.13	6.43
27	Scientific debate	sci_debate	212,607	14,929	227,536	2.75	1.02	2.47
28	Popularisation or scientific discovery	sci_discovery	344,877	62,495	407,372	4.46	4.27	4.43
29	Questioning of climate science	sci_ skepticism	64,736	2,065	66,801	0.84	0.14	0.73
30	Defense of climate science	sci_defense	48,511	0	48,511	0.63	0.00	0.53
<i>Environmental Frame</i>								
31	Environmental Frame (Primary Category)	environmental_ frame	206,531	74,554	281,085	2.67	5.09	3.06
32	Loss of natural environments	env_habitat	107,509	62,541	170,050	1.39	4.27	1.85
33	Loss of fauna and flora	env_species	118,580	49,797	168,377	1.53	3.40	1.83
<i>Cultural Frame</i>								
34	Cultural Frame (Primary Category)	cultural_ frame	290,437	82,465	372,902	3.75	5.63	4.05
35	Artistic representation	cult_art	202,421	44,568	246,989	2.62	3.05	2.68
36	Difficulty to host cultural or sports events	cult_event_ impact	22,993	34,855	57,848	0.30	2.38	0.63
37	Loss of indigenous practices	cult_ indigenous	7,200	18,778	25,978	0.09	1.28	0.28
38	Carbon footprint of the cultural and sports sectors	cult_ footprint	13	82,382	82,395	0.00	5.63	0.90
PRIMARY CATEGORIES								
<i>Actors/Messengers</i>								
39	Presence of Messengers (Primary Category)	messenger	3,843,269	644,020	4,487,289	49.68	44.00	48.78

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Supplementary Table S8 – Continued from previous page

#	Category	Code	Count			Proportion (%)		
			EN	FR	ALL	EN	FR	ALL
40	Health expert	msg_health	47,074	7,428	54,502	0.61	0.51	0.59
41	Economic expert	msg_economic	215,924	73,343	289,267	2.79	5.01	3.14
42	Security expert	msg_security	19	8,916	8,935	0.00	0.61	0.10
43	Legal expert	msg_legal	7	5,025	5,032	0.00	0.34	0.05
44	Cultural or Sport expert	msg_cultural	123,890	18,037	141,927	1.60	1.23	1.54
45	Natural scientist	msg_scientist	279,621	70,656	350,277	3.61	4.83	3.81
46	Social scientist	msg_social	32,227	19,609	51,836	0.42	1.34	0.56
47	Activist	msg_activist	96,990	25,189	122,179	1.25	1.72	1.33
48	Public official	msg_official	1,285,694	223,649	1,509,343	16.62	15.28	16.41
<i>Events</i>								
49	Presence of Events (Primary Category)	event	1,419,146	277,560	1,696,706	18.35	18.96	18.44
50	Extreme meteorological event	evt_weather	199,964	38,746	238,710	2.59	2.65	2.59
51	Meeting	evt_meeting	396,195	94,979	491,174	5.12	6.49	5.34
52	Publication	evt_publication	234,021	72,729	306,750	3.03	4.97	3.33
53	Election	evt_election	64,641	28,706	93,347	0.84	1.96	1.01
54	New policy	evt_policy	379,574	42,444	422,018	4.91	2.90	4.59
55	Judiciary decision	evt_judiciary	37,007	6,285	43,292	0.48	0.43	0.47
56	Cultural or Sports event	evt_cultural	964	22,998	23,962	0.01	1.57	0.26
57	Protest	evt_protest	7,728	5,293	13,021	0.10	0.36	0.14
<i>Solutions</i>								
58	Presence of Solutions (Primary Category)	solution	1,818,537	307,710	2,126,247	23.51	21.02	23.11
59	Mitigation strategy	sol_mitigation	1,258,602	201,352	1,459,954	16.27	13.76	15.87
60	Adaptation strategy	sol_adaptation	48,944	18,119	67,063	0.63	1.24	0.73

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Supplementary Table S8 – Continued from previous page

#	Category	Code	Count			Proportion (%)		
			EN	FR	ALL	EN	FR	ALL
EMOTIONAL TONE								
61	Emotion: positive	tone_positive	633,227	184,697	817,924	8.19	12.62	8.89
62	Emotion: negative	tone_negative	2,971,135	513,870	3,485,005	38.41	35.11	37.88
63	Emotion: neutral	tone_neutral	4,872,522	1,057,425	5,929,947	62.99	72.25	64.46
GEOGRAPHIC FOCUS								
64	Mention of Canada	canada	3,383,427	573,263	3,956,690	43.74	39.17	43.01
URGENCY TO ACT								
65	Urgency to act	urgency	160,053	35,784	195,837	2.07	2.44	2.13
Total Sentences			7,735,377	1,463,581	9,198,958	100.00	100.00	100.00

Note. Counts and proportions are computed live from CCF_Database.CCF_processed_data at table-build time; the “ALL” columns aggregate English and French. Proportions are expressed in percent of the total number of sentences in each language. The total sentence counts at the bottom of the table match the data-dictionary entry for CCF_processed_data in Supplementary Table S12.

Supplementary Table S9. Per-category inter-coder reliability metrics on the 1,000-sentence gold standard, with the blind phase (sentences 601–1000) reported separately.

#	Category	Code	Cohen’s κ (full)	Blind phase (sentences 601–1000)				n
				κ	Krippendorff α	Gwet AC1	% agree	
THEMATIC FRAMES								
Economic Frame								
1	Economic Frame (Primary Category)	economic_frame	0.605	0.625	0.625	0.735	0.868	400
2	Negative impacts of climate change on the economy	eco_neg_impact	0.694	0.542	0.542	0.915	0.958	400
3	Positive impacts of climate change on the economy	eco_pos_impact	0.928	0.831	0.831	0.990	0.995	400
4	Economic disadvantages of climate action	eco_cost	0.501	0.471	0.472	0.890	0.945	400
5	Economic benefits of climate action	eco_benefit	0.462	0.436	0.433	0.905	0.953	400
6	Carbon footprint of the economic sector	eco_footprint	0.354	0.441	0.437	0.820	0.910	400
Health Frame								
7	Health Frame (Primary Category)	health_frame	0.797	0.865	0.865	0.975	0.988	400
8	Negative impacts of climate change on health	health_neg_impact	0.790	0.851	0.851	0.975	0.988	400
9	Health co-benefits of climate action	health_cobenefit	0.304	-0.004	-0.004	0.980	0.990	400
Security Frame								
10	Security Frame (Primary Category)	security_frame	0.777	0.851	0.851	0.975	0.988	400
11	Presence of climate refugees	security_refugees	0.826	0.907	0.907	0.990	0.995	400

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#	Category	Code	Cohen's κ (full)	Blind phase (sentences 601–1000)				n
				κ	Krippendorff α	Gwet AC1	% agree	
12	Conflict	security_conflict	0.844	1.000	1.000	1.000	1.000	400
13	Post-disaster military assistance	security_military	0.283	-0.003	-0.001	0.990	0.995	400
14	Disruption of military operations	security_disruption	0.000	–	1.000	1.000	1.000	400
<i>Justice Frame</i>								
15	Justice Frame (Primary Category)	justice_frame	0.638	0.720	0.719	0.890	0.945	400
16	Winners and losers of climate action	justice_winners	0.275	0.386	0.381	0.925	0.963	400
17	Differentiated responsibility	justice_responsibility	0.637	0.655	0.654	0.950	0.975	400
18	Unequal vulnerability to climate change	justice_vulnerability	0.676	0.672	0.672	0.945	0.973	400
19	Unequal access to climate action	justice_access	0.058	0.168	0.160	0.910	0.955	400
20	Intergenerational justice	justice_intergen	0.645	0.731	0.731	0.975	0.988	400
<i>Political Frame</i>								
21	Political Frame (Primary Category)	political_frame	0.632	0.664	0.662	0.700	0.850	400
22	Policy action	pol_action	0.445	0.341	0.340	0.825	0.912	400
23	Political debate	pol_debate	0.498	0.542	0.534	0.615	0.807	400
24	Political positioning	pol_position	0.248	0.191	0.187	0.890	0.945	400
25	Public opinion data	pol_opinion	0.709	0.855	0.855	0.990	0.995	400
<i>Scientific Frame</i>								
26	Scientific Frame (Primary Category)	scientific_frame	0.579	0.564	0.562	0.770	0.885	400
27	Scientific debate	sci_debate	0.318	0.402	0.393	0.890	0.945	400

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Supplementary Table S9 – Continued from previous page

#	Category	Code	Cohen's κ (full)	Blind phase (sentences 601–1000)				n
				κ	Krippendorff α	Gwet AC1	% agree	
28	Popularisation or scientific discovery	sci_discovery	0.506	0.518	0.517	0.805	0.902	400
29	Questioning of climate science	sci_skepticism	0.290	0.196	0.191	0.960	0.980	400
30	Defense of climate science	sci_defense	0.661	0.748	0.748	0.990	0.995	400
<i>Environmental Frame</i>								
31	Environmental Frame (Primary Category)	environmental_frame	0.505	0.588	0.587	0.905	0.953	400
32	Loss of natural environments	env_habitat	0.358	0.450	0.448	0.910	0.955	400
33	Loss of fauna and flora	env_species	0.410	0.432	0.430	0.915	0.958	400
<i>Cultural Frame</i>								
34	Cultural Frame (Primary Category)	cultural_frame	0.594	0.757	0.758	0.955	0.978	400
35	Artistic representation	cult_art	0.696	0.792	0.792	0.970	0.985	400
36	Difficulty to host cultural or sports events	cult_event_impact	0.000	0.000	0.993	0.985	0.993	400
37	Loss of indigenous practices	cult_indigenous	0.665	1.000	1.000	1.000	1.000	400
38	Carbon footprint of the cultural and sports sectors	cult_footprint	0.666	–	1.000	1.000	1.000	400
PRIMARY CATEGORIES								
<i>Actors/Messengers</i>								
39	Presence of Messengers (Primary Category)	messenger	0.712	0.801	0.801	0.810	0.905	400
40	Health expert	msg_health	0.802	0.762	0.762	0.970	0.985	400
41	Economic expert	msg_economic	0.680	0.652	0.653	0.905	0.953	400
42	Security expert	msg_security	0.490	0.460	0.460	0.955	0.978	400
43	Legal expert	msg_legal	0.686	0.886	0.886	0.990	0.995	400
44	Cultural or Sport expert	msg_cultural	0.666	0.699	0.699	0.940	0.970	400
45	Natural scientist	msg_scientist	0.607	0.692	0.691	0.885	0.943	400

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Supplementary Table S9 – Continued from previous page

#	Category		Code	Cohen's κ (full)	Blind phase (sentences 601–1000)				n
					κ	Krippendorff α	Gwet AC1	% agree	
46	Social scientist		msg_social	0.568	0.653	0.653	0.945	0.973	400
47	Activist		msg_activist	0.621	0.560	0.561	0.895	0.948	400
48	Public official		msg_official	0.609	0.636	0.636	0.755	0.877	400
Events									
49	Presence of Events (Primary Category)		event	0.667	0.626	0.626	0.630	0.815	400
50	Extreme event	meteorological	evt_weather	0.700	0.718	0.718	0.945	0.973	400
51	Meeting		evt_meeting	0.669	0.762	0.762	0.925	0.963	400
52	Publication		evt_publication	0.778	0.807	0.807	0.925	0.963	400
53	Election		evt_election	0.732	0.678	0.678	0.955	0.978	400
54	New policy		evt_policy	0.502	0.398	0.397	0.785	0.892	400
55	Judiciary decision		evt_judiciary	0.810	0.828	0.828	0.980	0.990	400
56	Cultural or Sports event		evt_cultural	0.737	0.632	0.631	0.945	0.973	400
57	Protest		evt_protest	0.749	0.805	0.805	0.970	0.985	400
Solutions									
58	Presence of Solutions (Primary Category)		solution	0.699	0.650	0.649	0.785	0.892	400
59	Mitigation strategy		sol_mitigation	0.712	0.661	0.661	0.820	0.910	400
60	Adaptation strategy		sol_adaptation	0.636	0.556	0.556	0.940	0.970	400
EMOTIONAL TONE									
61	Emotion: positive		tone_positive	0.454	0.468	0.468	0.830	0.915	400
62	Emotion: negative		tone_negative	0.543	0.561	0.561	0.590	0.795	400
63	Emotion: neutral		tone_neutral	0.441	0.468	0.468	0.470	0.735	400
GEOGRAPHIC FOCUS									
64	Mention of Canada		canada	0.876	0.870	0.870	0.870	0.935	400
URGENCY TO ACT									
65	Urgency to act		urgency	0.176	0.083	0.061	0.895	0.948	400

Note. Cohen’s κ , Krippendorff’s α , and Gwet’s AC1 are computed between the expert annotator (gold standard) and the independent second annotator. The full-sample κ uses all 1,000 doubly-coded sentences; the blind-phase metrics restrict to sentences 601–1000, after which the second annotator coded without any communication with the expert. Categories with “–” in the blind-phase columns had no positive examples in sentences 601–1000.

Supplementary Table S10. Reliability tier assignment for each of the 65 annotation categories.

#	Category	Code	Validation F_1 macro			Blind κ	Tier			Flag
			All	EN	FR		All	EN	FR	
THEMATIC FRAMES										
Economic Frame										
1	Economic Frame (Primary Category)	economic_frame	0.830	0.847	0.814	0.625	A	A	A	
2	Negative impacts of climate change on the economy	eco_neg_impact	0.783	0.820	0.741	0.542	B	B	B	
3	Positive impacts of climate change on the economy	eco_pos_impact	0.684	0.806	0.508	0.831	B	A	C	
4	Economic disadvantages of climate action	eco_cost	0.723	0.741	0.702	0.471	B	B	B	
5	Economic benefits of climate action	eco_benefit	0.848	0.885	0.817	0.436	B	B	B	
6	Carbon footprint of the economic sector	eco_footprint	0.787	0.785	0.788	0.441	B	B	B	
Health Frame										
7	Health Frame (Primary Category)	health_frame	0.808	0.800	0.817	0.865	A	A	A	
8	Negative impacts of climate change on health	health_neg_impact	0.364	0.386	0.340	0.851	C	C	C	
9	Health co-benefits of climate action	health_cobenefit	0.186	0.252	0.103	-0.004	C	C	C	
Security Frame										
10	Security Frame (Primary Category)	security_frame	0.782	0.825	0.730	0.851	B	A	B	
11	Presence of climate refugees	security_refugees	0.481	0.476	0.417	0.907	C	C	C	

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Supplementary Table S10 – Continued from previous page

#	Category	Code	Validation F_1 macro			Blind κ	Tier			Flag
			All	EN	FR		All	EN	FR	
12	Conflict	security_conflict	0.324	0.358	0.286	1.000	C	C	C	
13	Post-disaster military assistance	security_military	0.045	0.000	0.045	-0.003	C	C	C	excluded
14	Disruption of military operations	security_disruption	1.000	0.000	1.000	–	C	C	C	excluded
<i>Justice Frame</i>										
15	Justice Frame (Primary Category)	justice_frame	0.857	0.832	0.880	0.720	A	A	A	
16	Winners and losers of climate action	justice_winners	0.708	0.654	0.737	0.386	C	C	C	PIRD*
17	Differentiated responsibility	justice_responsibility	0.860	0.900	0.823	0.655	A	A	A	
18	Unequal vulnerability to climate change	justice_vulnerability	0.845	0.786	0.894	0.672	A	B	A	
19	Unequal access to climate action	justice_access	0.651	0.475	0.796	0.168	C	C	C	
20	Intergenerational justice	justice_intergen	0.942	0.924	0.952	0.731	A	A	A	
<i>Political Frame</i>										
21	Political Frame (Primary Category)	political_frame	0.829	0.812	0.842	0.664	A	A	A	
22	Policy action	pol_action	0.756	0.802	0.722	0.341	C	C	C	PIRD*
23	Political debate	pol_debate	0.564	0.469	0.650	0.542	C	C	B	
24	Political positioning	pol_position	0.782	0.752	0.808	0.191	C	C	C	PIRD*
25	Public opinion data	pol_opinion	0.894	0.877	0.910	0.855	A	A	A	
<i>Scientific Frame</i>										
26	Scientific Frame (Primary Category)	scientific_frame	0.890	0.891	0.890	0.564	B	B	B	

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Supplementary Table S10 – Continued from previous page

#	Category	Code	Validation F_1 macro			Blind κ	Tier			Flag
			All	EN	FR		All	EN	FR	
27	Scientific debate	sci_debate	0.911	0.962	0.867	0.402	B	B	B	PIRD*
28	Popularisation or scientific discovery	sci_discovery	0.838	0.884	0.797	0.518	B	B	B	
29	Questioning of climate science	sci_skepticism	0.775	0.783	0.768	0.196	C	C	C	
30	Defense of climate science	sci_defense	0.682	0.769	0.460	0.748	B	B	C	
Environmental Frame										
31	Environmental Frame (Primary Category)	environmental_frame	0.871	0.866	0.875	0.588	B	B	B	
32	Loss of natural environments	env_habitat	0.721	0.804	0.573	0.450	B	B	C	
33	Loss of fauna and flora	env_species	0.703	0.714	0.692	0.432	B	B	B	
Cultural Frame										
34	Cultural Frame (Primary Category)	cultural_frame	0.708	0.802	0.581	0.757	B	A	C	
35	Artistic representation	cult_art	0.659	0.834	0.469	0.792	B	A	C	
36	Difficulty to host cultural or sports events	cult_event_impact	0.484	0.631	0.354	0.000	C	C	C	
37	Loss of indigenous practices	cult_indigenous	0.595	0.824	0.415	1.000	C	A	C	
38	Carbon footprint of the cultural and sports sectors	cult_footprint	0.351	0.702	0.014	–	C	B	C	
PRIMARY CATEGORIES										
Actors/Messengers										
39	Presence of Messengers (Primary Category)	messenger	0.961	0.955	0.967	0.801	A	A	A	
40	Health expert	msg_health	0.907	0.862	0.951	0.762	A	A	A	
41	Economic expert	msg_economic	0.874	0.847	0.895	0.652	A	A	A	

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Supplementary Table S10 – Continued from previous page

#	Category	Code	Validation F_1 macro			Blind κ	Tier			Flag
			All	EN	FR		All	EN	FR	
42	Security expert	msg_security	0.787	0.728	0.813	0.460	B	B	B	
43	Legal expert	msg_legal	0.914	0.861	0.927	0.886	A	A	A	
44	Cultural or Sport expert	msg_cultural	0.843	0.853	0.825	0.699	A	A	A	
45	Natural scientist	msg_scientist	0.907	0.920	0.896	0.692	A	A	A	
46	Social scientist	msg_social	0.944	0.938	0.948	0.653	A	A	A	
47	Activist	msg_activist	0.918	0.934	0.903	0.560	B	B	B	
48	Public official	msg_official	0.929	0.927	0.930	0.636	A	A	A	
Events										
49	Presence of Events (Primary Category)	event	0.897	0.881	0.912	0.626	A	A	A	PIRD*
50	Extreme meteorological event	evt_weather	0.889	0.887	0.891	0.718	A	A	A	
51	Meeting	evt_meeting	0.877	0.855	0.891	0.762	A	A	A	
52	Publication	evt_publication	0.958	0.950	0.962	0.807	A	A	A	
53	Election	evt_election	0.903	0.911	0.895	0.678	A	A	A	
54	New policy	evt_policy	0.829	0.821	0.837	0.398	C	C	C	
55	Judiciary decision	evt_judiciary	0.808	0.628	0.939	0.828	A	B	A	
56	Cultural or Sports event	evt_cultural	0.881	0.906	0.857	0.632	A	A	A	
57	Protest	evt_protest	0.943	0.982	0.909	0.805	A	A	A	
Solutions										
58	Presence of Solutions (Primary Category)	solution	0.900	0.891	0.909	0.650	A	A	A	
59	Mitigation strategy	sol_mitigation	0.839	0.872	0.801	0.661	A	A	A	
60	Adaptation strategy	sol_adaptation	0.895	0.889	0.903	0.556	B	B	B	
EMOTIONAL TONE										
61	Emotion: positive	tone_positive	0.702	0.712	0.692	0.468	B	B	B	
62	Emotion: negative	tone_negative	0.799	0.793	0.804	0.561	B	B	B	
63	Emotion: neutral	tone_neutral	0.777	0.756	0.798	0.468	B	B	B	

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Supplementary Table S10 – Continued from previous page

#	Category	Code	Validation F_1 macro			Blind κ	Tier			Flag
			All	EN	FR		All	EN	FR	
GEOGRAPHIC FOCUS										
64	Mention of Canada	canada	0.982	0.986	0.978	0.870	A	A	A	
URGENCY TO ACT										
65	Urgency to act	urgency	0.853	0.874	0.835	0.083	C	C	C	PIRD*
TIER COUNTS: All: A=27 / B=21 / C=17; EN: A=30 / B=20 / C=15; FR: A=27 / B=17 / C=21; PIRD*: 6; excluded from training: 2.										

Note. Each category is assigned to a reliability tier based on its validation F_1 macro and the blind-phase Cohen’s κ . **Tier A** (research-grade) requires $F_1 \geq 0.80$ and $\kappa \geq 0.60$; **Tier C** (exploratory) corresponds to $F_1 < 0.60$ or $\kappa < 0.40$; **Tier B** (use with caution) captures everything in between. The acronym **PIRD*** in the Flag column stands for *prevalence-induced reliability deflation*: it identifies categories with $F_1 \geq 0.70$ but blind-phase Cohen’s $\kappa < 0.40$, where the chance-corrected agreement is depressed by very low prevalence rather than by systematic disagreement between coders [18, 19]. The two categories excluded from training due to insufficient positive examples are flagged “excluded” and forced to Tier C in every language column.

Supplementary Table S11. Per-model training provenance: best epoch, training phase, validation macro F_1 , and positive-class support for every BERT/CamemBERT classifier.

#	Category	Code	English (BERT)				French (CamemBERT)			
			Epoch	Training phase	F_1	n^+	Epoch	Training phase	F_1	n^+
THEMATIC FRAMES										
Economic Frame										
1	Economic Frame (Primary Category)	economic_frame	4	r	0.845	24	13	n	0.885	28
2	Negative impacts of climate change on the economy	eco_neg_impact	12	n	0.821	6	3	n	0.917	8
3	Positive impacts of climate change on the economy	eco_pos_impact	6	n	0.664	5	4	r	0.698	3
4	Economic disadvantages of climate action	eco_cost	5	n	0.732	6	1	r	0.707	4
5	Economic benefits of climate action	eco_benefit	9	n	0.726	3	17	n	0.749	16
6	Carbon footprint of the economic sector	eco_footprint	8	n	0.897	7	16	n	0.904	8
Health Frame										
7	Health Frame (Primary Category)	health_frame	9	n	0.894	6	6	n	0.830	4
8	Negative impacts of climate change on health	health_neg_impact	1	n	0.455	5	1	n	0.429	3
9	Health co-benefits of climate action	health_cobenefit	17	r	0.486	1	2	r	0.119	2
Security Frame										
10	Security Frame (Primary Category)	security_frame	9	n	0.933	12	11	n	0.898	2
11	Presence of climate refugees	security_refugees	2	n	1.000	1	9	r	1.000	1
12	Conflict	security_conflict	1	n	1.000	1	14	r	1.000	1
13	Post-disaster military assistance	security_military	—	—	—	—	3	r	0.026	2
14	Disruption of military operations	security_disruption	—	—	—	—	18	n	1.000	1
Justice Frame										
15	Justice Frame (Primary Category)	justice_frame	5	n	0.847	32	10	n	0.849	30

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Supplementary Table S11 – Continued from previous page

#	Category	Code	English (BERT)				French (CamemBERT)			
			Epoch	Training phase	F_1	n^+	Epoch	Training phase	F_1	n^+
16	Winners and losers of climate action	justice_winners	13	n	0.800	2	14	n	0.795	2
17	Differentiated responsibility	justice_responsibility	4	n	0.883	4	3	n	0.750	4
18	Unequal vulnerability to climate change	justice_vulnerability	19	r	0.782	6	13	n	0.798	5
19	Unequal access to climate action	justice_access	13	r	0.182	2	4	r	0.809	9
20	Intergenerational justice	justice_intergen	9	n	0.966	7	11	n	0.899	3
<i>Political Frame</i>										
21	Political Frame (Primary Category)	political_frame	3	n	0.853	46	5	n	0.831	133
22	Policy action	pol_action	2	n	0.803	17	4	n	0.816	6
23	Political debate	pol_debate	7	n	0.569	38	4	n	0.768	43
24	Political positioning	pol_position	5	n	0.848	6	4	n	0.895	3
25	Public opinion data	pol_opinion	4	n	0.954	6	12	n	1.000	2
<i>Scientific Frame</i>										
26	Scientific Frame (Primary Category)	scientific_frame	3	n	0.869	27	5	n	0.832	56
27	Scientific debate	sci_debate	5	n	0.793	10	7	n	0.688	5
28	Popularisation or scientific discovery	sci_discovery	3	n	0.814	14	6	n	0.844	16
29	Questioning of climate science	sci_skepticism	3	r	0.827	5	5	r	0.664	4
30	Defense of climate science	sci_defense	6	n	0.752	4	1	n	0.459	3
<i>Environmental Frame</i>										
31	Environmental Frame (Primary Category)	environmental_frame	7	n	0.915	9	13	r	0.802	6
32	Loss of natural environments	env_habitat	7	n	1.000	5	5	n	0.829	3
33	Loss of fauna and flora	env_species	5	n	0.873	4	3	n	0.829	3
<i>Cultural Frame</i>										
34	Cultural Frame (Primary Category)	cultural_frame	7	n	0.879	24	13	r	0.913	5
35	Artistic representation	cult_art	2	n	1.000	2	7	n	1.000	2
36	Difficulty to host cultural or sports events	cult_event_impact	18	n	0.850	7	4	n	1.000	2

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#	Category	Code	English (BERT)				French (CamemBERT)			
			Epoch	Training phase	F_1	n^+	Epoch	Training phase	F_1	n^+
37	Loss of indigenous practices	cult_indigenous	6	n	1.000	1	3	r	0.462	2
38	Carbon footprint of the cultural and sports sectors	cult_footprint	7	n	1.000	1	3	r	0.074	1
PRIMARY CATEGORIES										
<i>Actors/Messengers</i>										
39	Presence of Messengers (Primary Category)	messenger	11	n	0.908	72	6	n	0.922	227
40	Health expert	msg_health	9	r	0.927	6	8	n	0.954	6
41	Economic expert	msg_economic	4	n	0.785	6	3	r	0.861	7
42	Security expert	msg_security	8	n	0.830	2	9	n	1.000	1
43	Legal expert	msg_legal	4	n	0.833	2	11	n	0.883	6
44	Cultural or Sport expert	msg_cultural	3	n	1.000	1	3	r	0.773	9
45	Natural scientist	msg_scientist	8	n	0.883	37	8	n	0.902	12
46	Social scientist	msg_social	7	n	0.883	7	16	r	0.816	12
47	Activist	msg_activist	5	n	0.775	5	5	n	1.000	6
48	Public official	msg_official	4	n	0.800	18	10	n	0.943	24
<i>Events</i>										
49	Presence of Events (Primary Category)	event	11	n	0.865	33	10	n	0.876	113
50	Extreme meteorological event	evt_weather	3	n	1.000	8	7	n	0.912	6
51	Meeting	evt_meeting	8	n	0.880	8	3	n	0.969	11
52	Publication	evt_publication	4	n	0.931	25	7	n	1.000	12
53	Election	evt_election	4	n	1.000	1	10	r	0.899	5
54	New policy	evt_policy	7	n	0.856	6	8	n	0.841	6
55	Judiciary decision	evt_judiciary	7	n	1.000	1	9	n	1.000	1
56	Cultural or Sports event	evt_cultural	4	n	0.664	5	7	n	1.000	1
57	Protest	evt_protest	4	n	0.944	5	5	n	0.862	6
<i>Solutions</i>										
58	Presence of Solutions (Primary Category)	solution	6	n	0.825	90	4	n	0.911	46
59	Mitigation strategy	sol_mitigation	16	n	0.842	79	7	n	0.877	105
60	Adaptation strategy	sol_adaptation	9	n	0.843	15	10	n	0.894	21
EMOTIONAL TONE										
61	Emotion: positive	tone_positive	8	n	0.746	11	6	n	0.829	16
62	Emotion: negative	tone_negative	2	n	0.745	152	16	n	0.804	158

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#	Category	Code	English (BERT)				French (CamemBERT)			
			Epoch	Training phase	F_1	n^+	Epoch	Training phase	F_1	n^+
63	Emotion: neutral	tone_neutral	4	n	0.709	205	4	n	0.741	80
GEOGRAPHIC FOCUS										
64	Mention of Canada	canada	7	n	0.955	136	12	n	0.972	54
URGENCY TO ACT										
65	Urgency to act	urgency	10	r	0.783	20	7	n	0.816	19
Summary: 128 trained models. Reinforcement triggered for 45 models; best epoch retained from the reinforced phase for 23 models. Static training configuration (optimiser, learning rate, batch size, random seed, hardware) is reported in the Methods section.										

Note. The English models are BERT-base-uncased and the French models are CamemBERT-base, both trained through the AugmentedSocialScientist framework. The four per-language sub-columns are: **Ep.** (best epoch retained by the combined-criterion best-model selector $0.7 \cdot F_1^{(1)} + 0.3 \cdot \text{macro-}F_1$), **Ph.** (training phase — n for normal, r for reinforced training, the latter firing when the positive-class F_1 falls below 0.60 during normal training), F_1 (macro F_1 at the retained epoch), and n^+ (number of positive validation examples; “–” denotes models that failed to train because of insufficient positive examples). The static configuration (optimiser AdamW with $\epsilon = 10^{-8}$, learning rate 5×10^{-5} in normal training and 1×10^{-5} in the reinforced phase, batch size 32 normal / 64 reinforced, maximum sequence length 512 tokens, up to 20 epochs in each phase, random seed 42 for Python/NumPy/PyTorch RNGs, linear schedule with zero warm-up, weighted cross-entropy in normal training and a **WeightedRandomSampler** in the reinforced phase, training on an Apple Mac Studio M2 Ultra with 128 GB unified memory) is documented in full in the Methods section of the main manuscript.

Supplementary Table S12. Complete data dictionary of the enriched CCF Database: every column of every table, its type, and a one-line definition.

Column	Type	Definition
Table CCF_full_data (<i>article-level metadata, 266,271 rows</i>)		
doc_id	BIGINT (PK)	Unique article identifier.
news_type	TEXT	Editorial category (news, opinion, editorial, etc.).
title	TEXT	Article headline.
author	TEXT	Author byline when available.
media	TEXT	Newspaper name.
words_count	BIGINT	Total word count of the article.
date	DATE	Publication date.
language	TEXT	“EN” or “FR”.
page_number	TEXT	Print page (free text, e.g. “A1” or “B3”).
Table CCF_processed_data (<i>sentence-level annotations, 9,198,958 rows</i>)		
doc_id	BIGINT	Foreign key to CCF_full_data.
sentence_id	BIGINT	Sentence position within the article (starting at 1; 0 reserved for title in the embeddings table).
<i>Ten article-level metadata columns are duplicated here for query convenience (news_type, title, author, media, words_count, date, language, page_number).</i>		
65 annotation columns	INTEGER (0/1, nullable)	One binary flag per category from Supplementary Table S3. Sub-categories are NULL when their parent primary detection is negative.
ner_entities	TEXT (JSON)	Per-sentence NER output as {"PER": [...], "ORG": [...], "LOC": [...]}.
Table CCF_article_aggregates (<i>article-level rollup, 266,271 rows</i>)		
doc_id	BIGINT (PK)	Foreign key to CCF_full_data.
n_sentences	INTEGER	Total number of two-sentence analytical units in the article.
n_sentences_en	INTEGER	Subset in English.
n_sentences_fr	INTEGER	Subset in French.
65 prop_X columns	DOUBLE PRECISION ([0, 1])	Article-level proportion of sentences positive for category <i>X</i> (sum / n_sentences, with NULL sub-categories coalesced to 0).
dominant_frame	TEXT	Thematic frame with the highest prop_X, or NULL for articles with no positive frame.
dominant_frame_prop	DOUBLE PRECISION	Proportion of the dominant frame.
entropy_frames	DOUBLE PRECISION	Shannon entropy (natural base) of the eight thematic-frame proportions; 0 when the article has no positive frame.

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Column	Type	Definition
n_unique_messenger_sub	SMALLINT	Number of distinct messenger sub-categories present in the article (0–9).
n_unique_event_sub	SMALLINT	Same for event sub-categories (0–8).
n_unique_solution_sub	SMALLINT	Same for solution sub-categories (0–2).
Table CCF_article_entities (<i>NER rollup, 266,271 rows</i>)		
doc_id	BIGINT (PK)	Foreign key to CCF_full_data.
entities_per	JSONB	Deduplicated array of persons mentioned in the article.
entities_org	JSONB	Deduplicated array of organisations.
entities_loc	JSONB	Deduplicated array of locations.
n_unique_per	INTEGER	Cardinality of <code>entities_per</code> .
n_unique_org	INTEGER	Cardinality of <code>entities_org</code> .
n_unique_loc	INTEGER	Cardinality of <code>entities_loc</code> .
n_sentences_with_per	INTEGER	Number of sentences with at least one PER entity.
n_sentences_with_org	INTEGER	Same for ORG.
n_sentences_with_loc	INTEGER	Same for LOC.
Table CCF_reliability_tiers (<i>per-category quality lookup, 65 rows</i>)		
number	SMALLINT (PK)	Table B1 number (1–65).
category	TEXT	Readable category name.
code	TEXT (UNIQUE)	Column code as used in CCF_processed_data.
tier_overall	CHAR(1)	Tier on the combined corpus: “A”, “B”, or “C”.
tier_en	CHAR(1)	Tier on the English subset.
tier_fr	CHAR(1)	Tier on the French subset.
paradox_kappa	BOOLEAN	True if the category exhibits prevalence-induced reliability deflation, i.e. $F_1 \geq 0.70$ and $\kappa < 0.40$. The column name is kept for backwards compatibility with the original schema.
exclusion_reason	TEXT	“insufficient_training_data” for the two categories that could not be trained, NULL otherwise.

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Column	Type	Definition
f1_macro_all, f1_macro_en, f1_macro_fr	DOUBLE PRECISION	Validation macro F_1 score on the gold standard.
support_all, support_en, support_fr	INTEGER	Number of positive (class 1) validation examples.
cohens_kappa_blind_phase, gwet_ac1_blind_phase, krippendorff_alpha_blind_phase	DOUBLE PRECISION	Inter-coder agreement on the blind phase (sentences 601–1000).
n_blind_phase	INTEGER	Sentence count of the blind phase.
prev_coder1_full, prev_coder2_full	DOUBLE PRECISION	Prevalence (proportion of positives) per coder on the full 1,000-sentence sample.
Table CCF_sentence_embeddings (<i>BGE-M3 dense vectors, 9,462,845 rows</i>)		
doc_id	BIGINT (PK, with sentence_id)	Foreign key to CCF_full_data.
sentence_id	INTEGER (PK)	Sentence position within the article. <code>sentence_id</code> = 0 is reserved for the article title.
embedding	halfvec(1024)	L2-normalised BGE-M3 vector. <code>pgvector</code> cosine distance via <code><=></code> ; HNSW index on cosine ops.

Note. All annotation column codes (`economic_frame`, `eco_neg_impact`, etc.) and their definitions appear in Supplementary Table S3.

Supplementary references

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